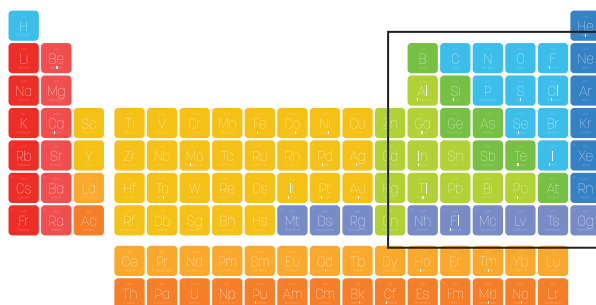


JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



A periodic table of elements is displayed within a white diamond-shaped frame. The elements are color-coded by groups: Group 1 (red), Group 2 (orange), Groups 3-10 (yellow), Groups 11-18 (green), and Groups 19-20 (blue). The lanthanide and actinide series are shown at the bottom of the table.

STUDY MATERIAL

Carboxylic Acid & its Derivatives

English Medium

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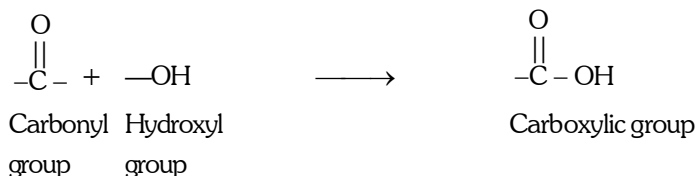
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CARBOXYLIC ACID AND ITS DERIVATIVES

1.0 INTRODUCTION - CARBOXYLIC ACID

Organic compounds having $-\text{COOH}$ group are called as Carboxylic acids. This functional group is composed of

Carbonyl ($\text{—}\overset{\text{O}}{\parallel}{\text{C}}\text{—}$) and hydroxyl (—OH) group.



The properties of the carboxylic group are not simply the combined properties of these two groups, but it has its own distinctive properties. The acidic nature of carboxylic acids is due to the presence of replaceable H-atom in the Carboxylic group. The general formula is $\text{C}_n\text{H}_{2n}\text{O}_2$.

1.1 CLASSIFICATION

1. Monocarboxylic acid (RCOOH)

Having one carboxylic group, also called monobasic acid. General formula - $\text{C}_n\text{H}_{2n}\text{O}_2$ ($n = 1, 2, 3, \dots$). Higher mono carboxylic acids are called **fatty acids**.

Example : CH_3COOH acetic acid

2. Dicarboxylic acid

Having two carboxylic groups, also called dibasic acid.

Example : $\begin{array}{c} \text{COOH} \\ | \\ \text{COOH} \end{array}$ Oxalic acid

3. Tricarboxylic acid

Having three carboxylic groups also called tribasic acid.

Example : $\begin{array}{c} \text{CH}_2\text{COOH} \\ | \\ \text{HO—C—COOH} \\ | \\ \text{CH}_2\text{COOH} \end{array}$ Citric acid

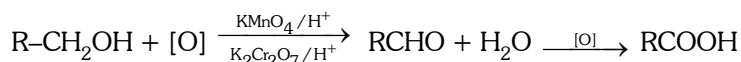
NOMENCLATURE

Acid	Common name	IUPAC name
HCOOH	Formic acid (formica-red ants)	Methanoic acid
CH_3COOH	Acetic acid (acetum-vinegar)	Ethanoic acid
$\text{CH}_3\text{CH}_2\text{COOH}$	Propionic acid (Propan-first pion-fat)	Propanoic acid
$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	Butyric acid (Butter-butyrum)	Butanoic acid
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$	Valeric acid (valerian-plant root)	Pentanoic acid
$\text{C}_5\text{H}_{11}\text{COOH}$	Caproic acid	Hexanoic acid
$\text{C}_7\text{H}_{15}\text{COOH}$	Caprylic acid	Octanoic acid
$\text{C}_9\text{H}_{19}\text{COOH}$	Capric acid	Decanoic acid

Last three acids are found in goat fat - (Caper-Goat).

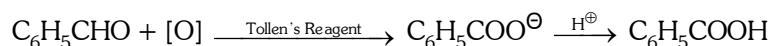
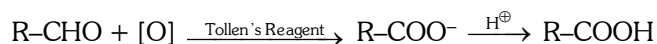
1.2 GENERAL METHODS OF PREPARATION

1. By oxidation of primary alcohol with acidic KMnO_4 or acidic $\text{K}_2\text{Cr}_2\text{O}_7$

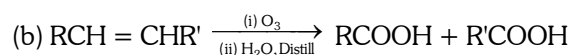
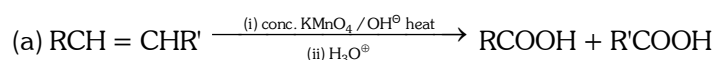


2. By oxidation of aldehydes

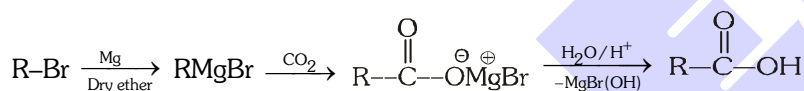
Aldehydes on oxidation with usual oxidizing agent gives carboxylic acid with same number of carbon atoms as in the aldehyde.



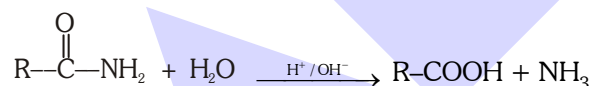
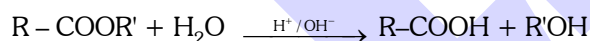
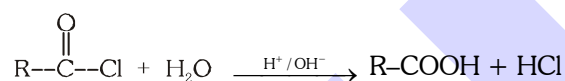
3. By oxidation of alkenes



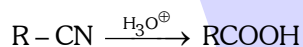
4. By Carboxylation of Grignard Reagent



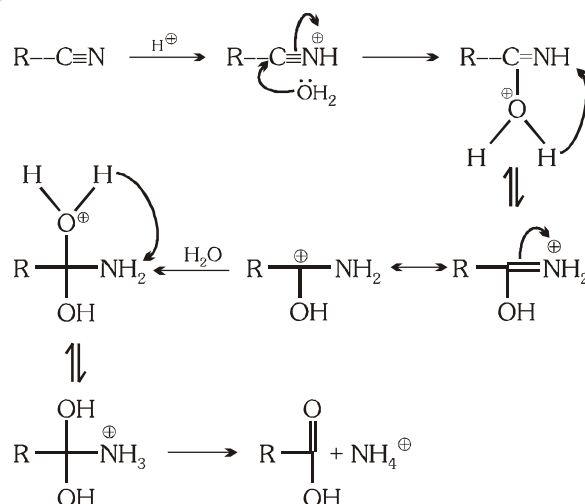
5. By hydrolysis of derivatives of carboxylic acid



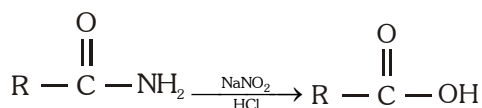
6. Cyanide hydrolysis with dilute acids



The mechanism of hydrolysis of R-CN is as follows.

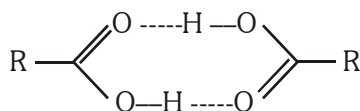


7. From Amides



1.3 PHYSICAL PROPERTIES OF CARBOXYLIC ACID

These are polar substances and can form H-bonds with each other to form dimer structures.



1. Boiling Point

Due to dimeric structure, the effective molecular mass of the acid becomes double the actual mass. Hence carboxylic acids have higher boiling points than alcohols of comparable molecular masses.

2. Solubility

Due to hydrogen bonding carboxylic acid show appreciable solubility in water. Its solubility in water is greater than alcohol because H-bonding strength is greater in carboxylic acid than alcohol.

3. Melting Point

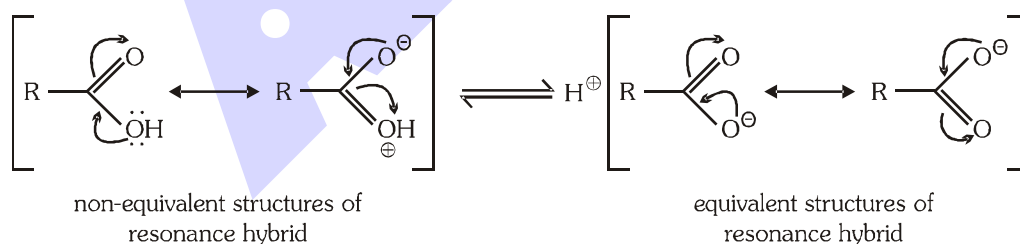
Melting point of the carboxylic acid with even number of carbon atoms is higher than acid with odd number of carbon atoms. Such effect is observed in first ten members of the homologous series. This feature is based on the fact that in the carboxylic acids with even number of carbon atoms, the terminal methyl group and carboxylic group are on the opposite sides of zig-zag carbon chain. Hence they fit better in the crystal lattice resulting in stronger inter molecular forces on the other hands acids with odd number of carbon atom have carboxyl and terminal methyl group on the same side of zig-zag carbon chain which result in poor fitting in the crystal lattice. This causes a weak forces among molecules and result for the relatively lower melting point.

The melting point are usually higher than those of aliphatic acid of comparable molecular masses. This is due to planar structure of benzene ring in the acid which can pack closely in the crystal than aliphatic acids.

1.4 CHEMICAL PROPERTIES OF CARBOXYLIC ACID

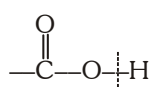
1. Acidity of carboxylic acid

Acidity is relative ease with which it loses a proton leaving behind the anion. Its acid strength depends upon the difference in the stability of the acid and its anion.

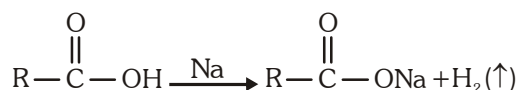


both acid and its anion are stabilized by resonance, stabilization is far greater for the anion than for acid because anion gives two identical resonating structure.

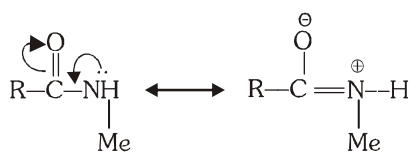
Reaction due to cleavage of —O—H bond as acid



• Reaction with active metals [alkali and alkaline metal]



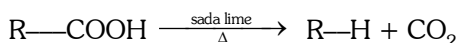
The lone pair electron on N atom undergoes resonance and hence N atom carries partial +ve charge on it.



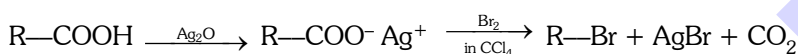
Thus extent of H-bonding increases which makes the boiling and melting points of amides higher than other acid derivatives.

6. Soda lime Decarboxylation

Carboxylic acid on heating with soda lime (NaOH and CaO) give alkane with one carbon atom less than the parent acid.



7. Hunsdicker Reaction



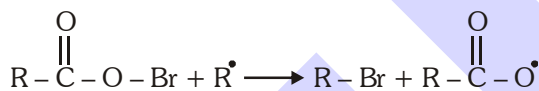
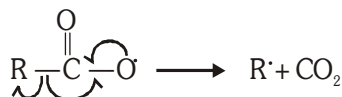
Mechanism

Reaction proceeds through free radical mechanism in various steps.

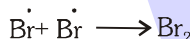
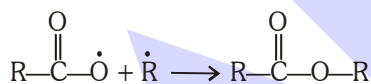
(i) Chain initiation :



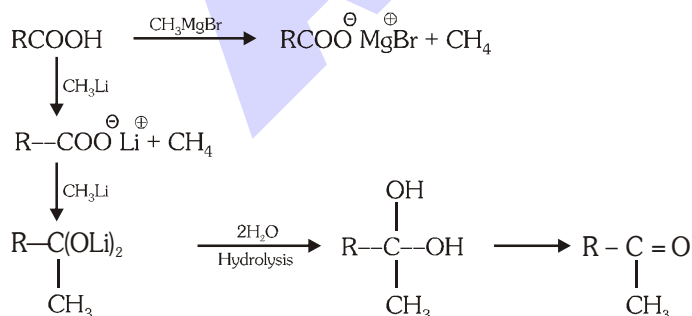
(ii) Chain propagation step :



(iii) Chain termination :



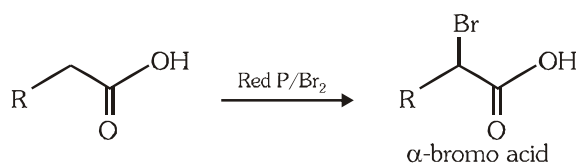
8. With excess of organometallic compounds

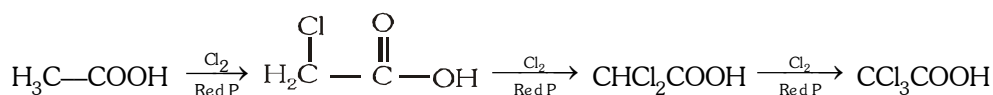


2 moles of organolithium compounds are needed to form ketone.

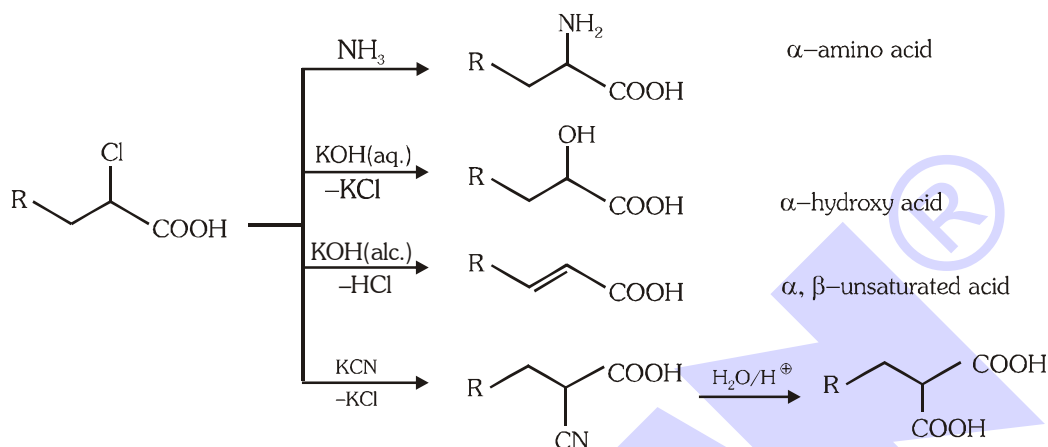
9. Hell-Volhard-Zelinsky Reaction

This is α -halogenation of a carboxylic acid.



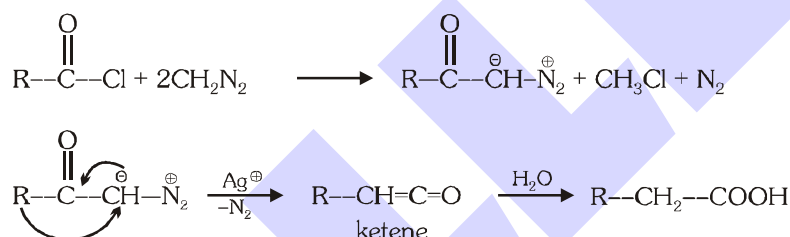


The reaction has a great synthetic importance as the halogen atom can be replaced by a number of other groups giving useful products.



10. Arndt Eistert Reaction

Reaction of acyl chloride with diazomethane in presence of Ag_2O and water. (Arndt – Eistert synthesis). This reaction convert acyl chloride to carboxylic acid with one carbon atom more :

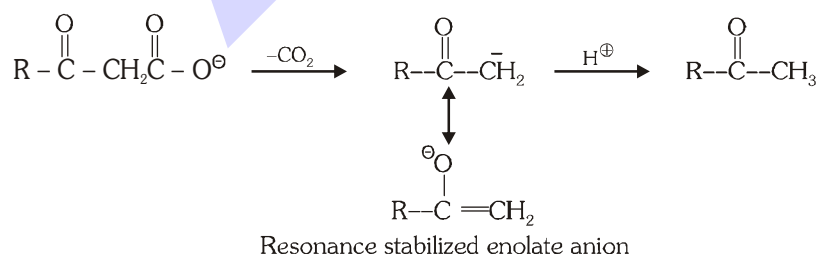


11. Heating of β -keto acid



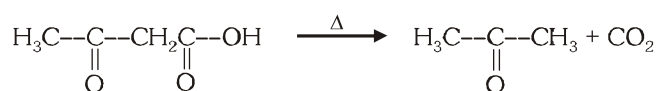
There are two facts on which ease of decarboxylation depends.

(i) When the carboxylate ion decarboxylates, it forms a resonance stabilized enolate ion.

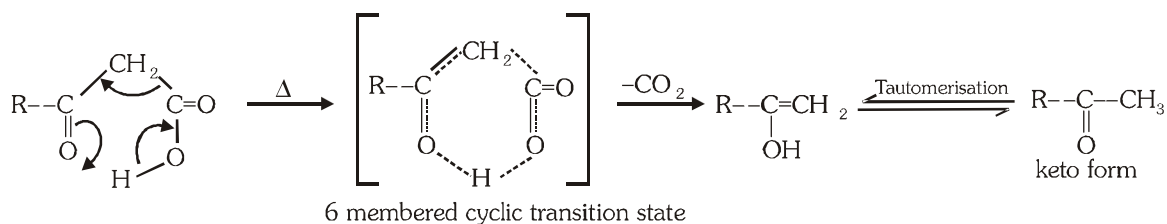


Due to the resonance enolate ion is much more stable than normal carbanion.

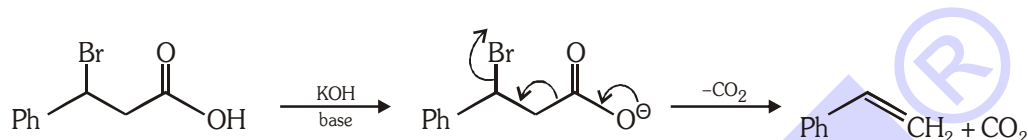
(ii) When the acid itself decarboxylates it can do so through a six-membered cyclic transition state.



• Mechanism

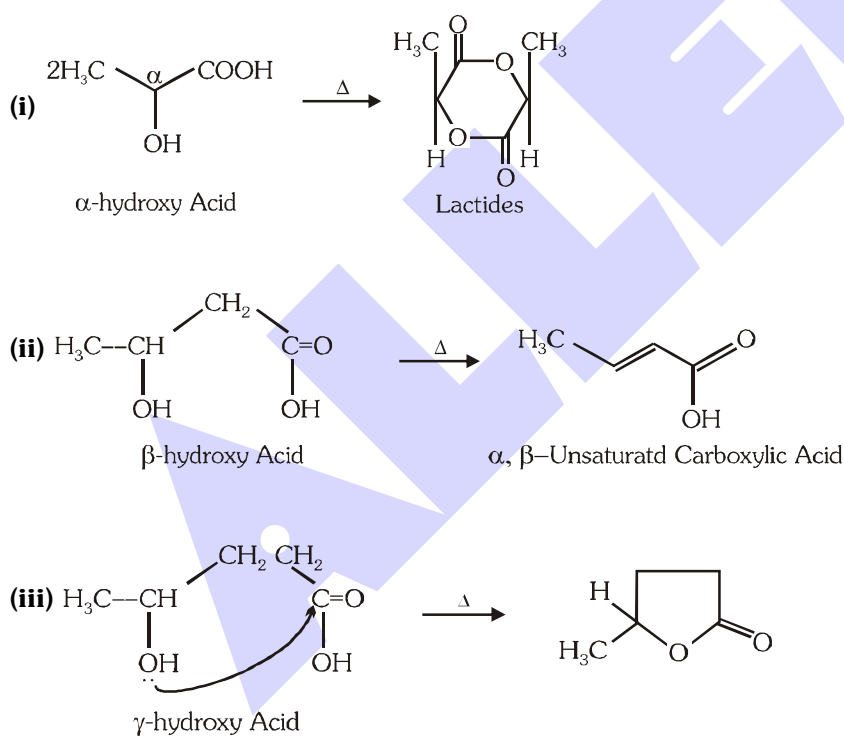


β-keto acid on warming in presence of a base undergoes rapid removal of CO₂.

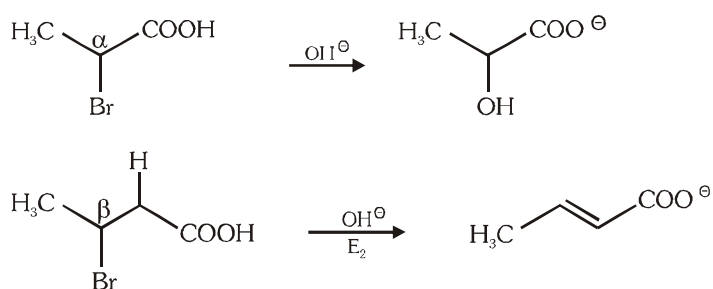


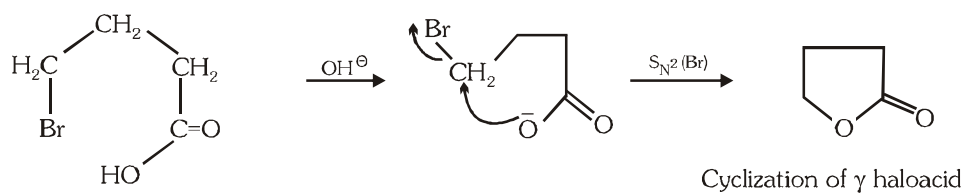
This decarboxylation proceeds through elimination.

12. Heating of α, β and γ Hydroxy Acid



13. Reaction of α, β and γ halo carboxylic acid with aq. NaOH



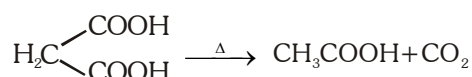


14. Heating of Dicarboxylic Acids

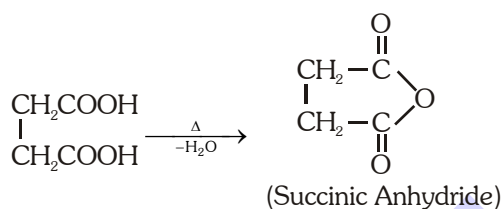
- 1, 2 - dicarboxylic acid**



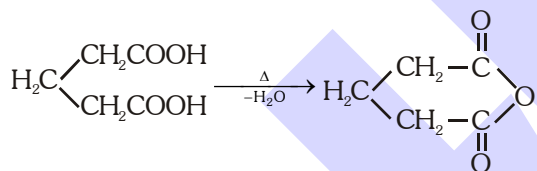
- 1, 3 - dicarboxylic acid**



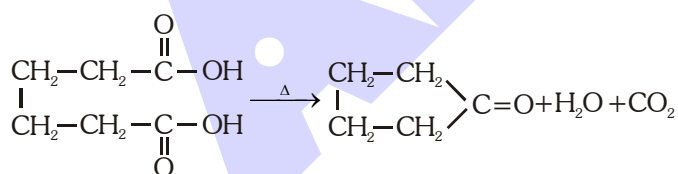
- 1, 4 - dicarboxylic acid**



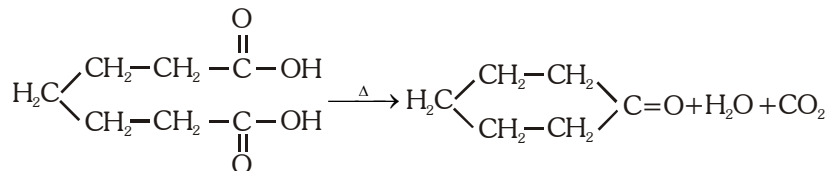
- 1, 5 - dicarboxylic acid**



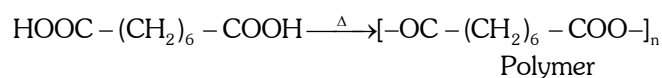
- 1, 6 - dicarboxylic acid**



- 1, 7 - dicarboxylic acid**



- 1, 8 - dicarboxylic acid**

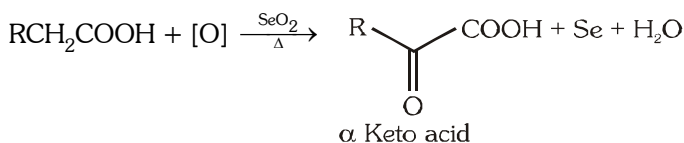


- Blanc's Rule**

(1,4), (1,5) dicarboxylic acids give cyclic anhydride whereas (1,6), (1,7) dicarboxylic acids give cyclic ketones on heating.

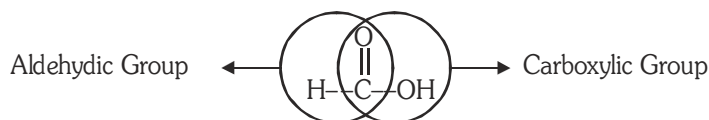
15. Oxidation of Carboxylic acid at α -position

Oxidation can also occur at α carbon atom on treatment with oxidizing agent like selenium dioxide SeO_2 .



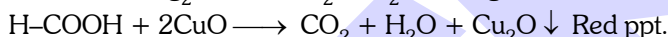
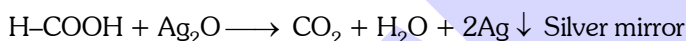
16. Abnormal Behaviour of Formic Acid

The behaviour of formic acid is different from other carboxylic acid due to presence of aldehydic group.

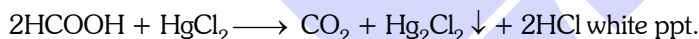


• Reaction with Tollen's Reagent and Fehling Solution

(a) **Tollen's reagents** : formic acid behaves as a reducing agent and reduces Tollen's reagent or Fehling solution. But others acid fail to do so.



(b) **Reaction with HgCl_2** : Formic acid forms white ppt. of Hg_2Cl_2 with HgCl_2 which is converted to Grey ppt of mercury.



Test for HCOOH and CH_3COOH

	Test	HCOOH	CH_3COOH
(i)	Reducing character	Reducing agents	
	Tollen reagent	Silver mirror	—
	Fehling solution	Cu_2O red	—
	HgCl_2	Hg_2Cl_2	—
	(Corrosive sublimate)	(Calomel)	—
	$\text{K}_2\text{Cr}_2\text{O}_7$	Cr^{+3}	—
(ii)	Decarboxylation.	$\text{Na}_2\text{CO}_3 + \text{H}_2$	CH_4

1.5 USES OF FORMIC ACID

- | | |
|---|--|
| <p>(i) As an antiseptic</p> <p>(iii) For leather tanning.</p> <p>(v) As a coagulating agent for rubber.</p> | <p>(ii) For preservation of fruits.</p> <p>(iv) In dyeing wool and cotton fabrics.</p> <p>(vi) For hydrogenation of oil as Ni-formate.</p> |
|---|--|

1.6 USES OF ACETIC ACID

- (i) Vinegar (6 - 10% solution) used as **table acid** and manufacture of pickles.
- (ii) In the form of salts, it is used in medicine and paints.
- (iii) For manufacture of rubber from latex and casein from milk CH_3COOH is used as coagulant.
- (iv) Al and Cr acetates are used as mordants.
- (v) In the manufacture of dyes and perfumes.
- (vi) As a solvent and laboratory reagent.

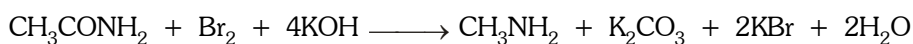
GOLDEN KEY POINTS

- Carboxylic acid gives CO_2 gas on reaction with NaHCO_3 .
- Carboxylic acid gives fruity smell on reaction with alcohol in acidic medium.

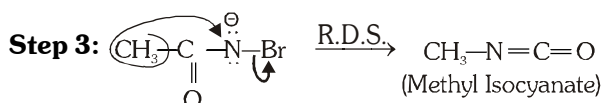
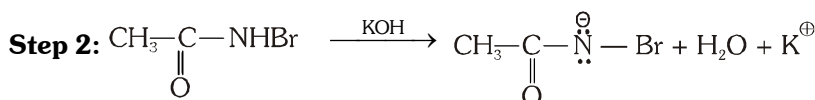
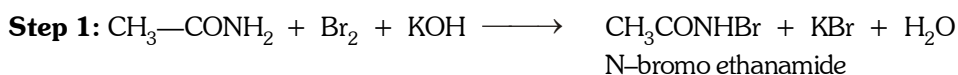
1.7 DEGRADATION REACTIONS

1. By Hofmann's Bromamide Reaction (Hofmann's Hypobromite Reaction)

This is a general method for the conversion of alkanamides in to one carbon less primary amines. Ethanamide is heated with bromine and excess of KOH.

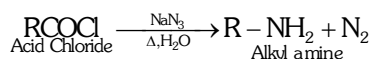


Mechanism

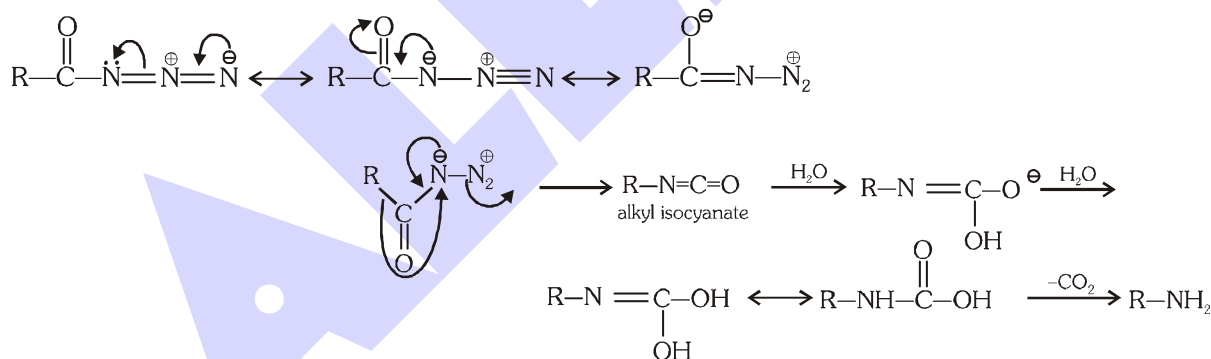
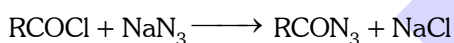


2. Curtius Reaction

Acid chloride on treatment with sodium azide give acid azides which on pyrolysis gives isocyanates which on hydrolysis gives corresponding amines.

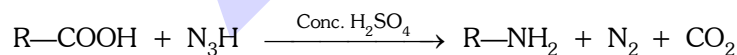


Mechanism

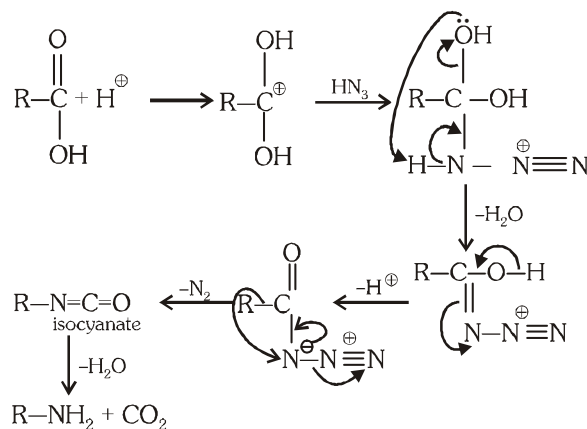


3. Schmidt Reaction

In presence of conc. H_2SO_4 alkanic acid reacts with hydrazoic acid (N_3H) to yield alkylamine.

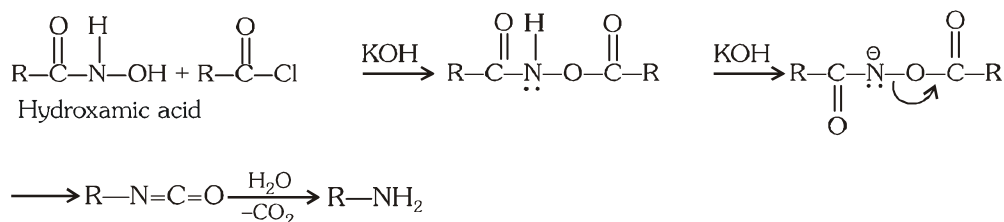


Mechanism



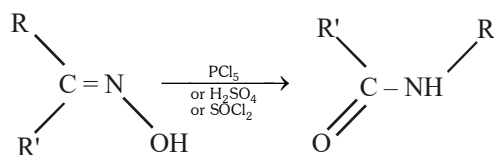
4. Lossen Rearrangement Reaction

In this reaction hydroxamic acid undergoes rearrangement and gives alkyl amine.

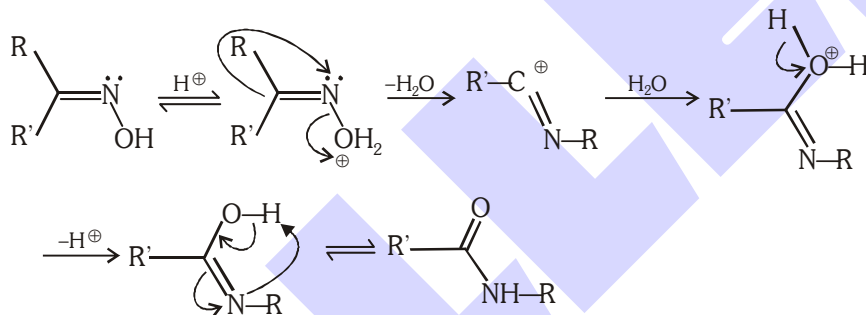


1.8 THE BECKMANN REARRANGEMENT

The acid catalysed transformation of a ketoxime to an N-substituted amide is known as the Beckmann rearrangement.



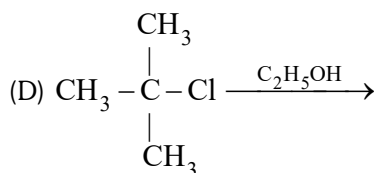
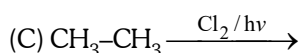
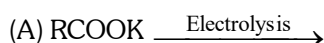
• Mechanism



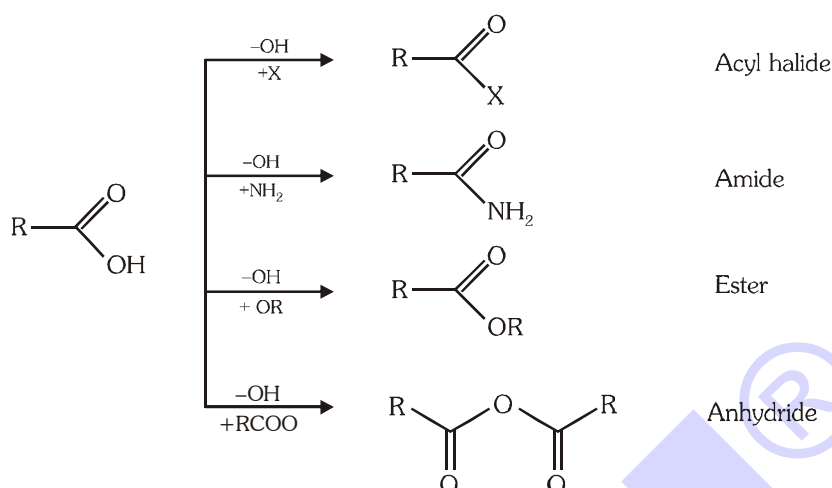
The rearrangement is catalysed by a variety of acidic reagents such as H_3PO_2 , H_2SO_4 , SOCl_2 , PCl_5 , etc.

BEGINNER'S BOX-1

- Which of the following orders of acid strength is correct :
 (A) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC}\equiv\text{CH}$ (B) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC}\equiv\text{CH}$
 (C) $\text{RCOOH} > \text{HOH} > \text{HC}\equiv\text{CH} > \text{ROH}$ (D) $\text{RCOOH} > \text{HC}\equiv\text{CH} > \text{HOH} > \text{ROH}$
- Which of the following compounds gives carbondioxide with NaHCO_3 ?
 (A) Acetic acid (B) Hexanol (C) Phenol (D) Acetylene
- In which reaction product is hydrocarbon?

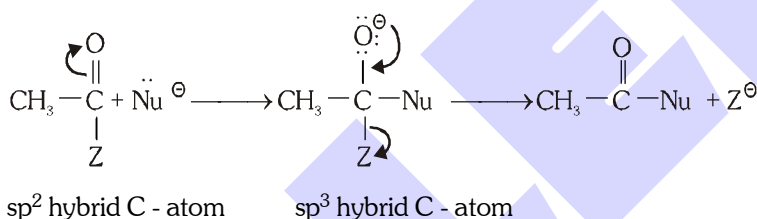


- Benzoic acid reacts with $\text{Ca}(\text{OH})_2$. The product obtained on dry distillation gives—
 (A) Benzophenone (B) Acetophenone (C) Benzaldehyde (D) None of these



2.0 CHARACTERISTIC REACTION FOR ACID DERIVATIVES IS NUCLEOPHILIC SUBSTITUTION REACTION

- **Mechanism**



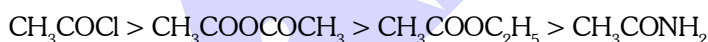
In this reaction Z is leaving group. Weak bases are good leaving groups.

Reactivity order - depends on the basic Character of Z

basicity : $\text{Cl}^- < \text{CH}_3\text{COO}^- < \text{C}_2\text{H}_5\text{O}^- < \text{NH}_2^-$

In the given groups Cl^- is the weakest base so it is the best leaving group.

- **Reactivity order**



In acid derivatives the carbonyl group $>\text{CO}$ is attached to highly electronegative $-\text{Cl}$, CH_3COO^- , $-\text{NH}_2$ etc. group due to electron withdrawing effect of these groups, the electron density on the carbonyl carbon is reduced.

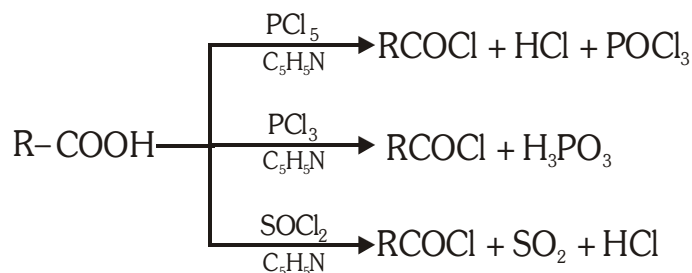
further. Thus acetyl group is readily attacked by $\ddot{\text{N}}_{\text{U}}$ shows nucleophilic substitution reaction.

3.0 ACYL CHLORIDE (RCOCl)

These are the derivatives of carboxylic acid in which hydroxyl ($-\text{OH}$) part of carboxyl group is replaced by halo group. The most reactive compound of halo leaving group is chloro compounds.

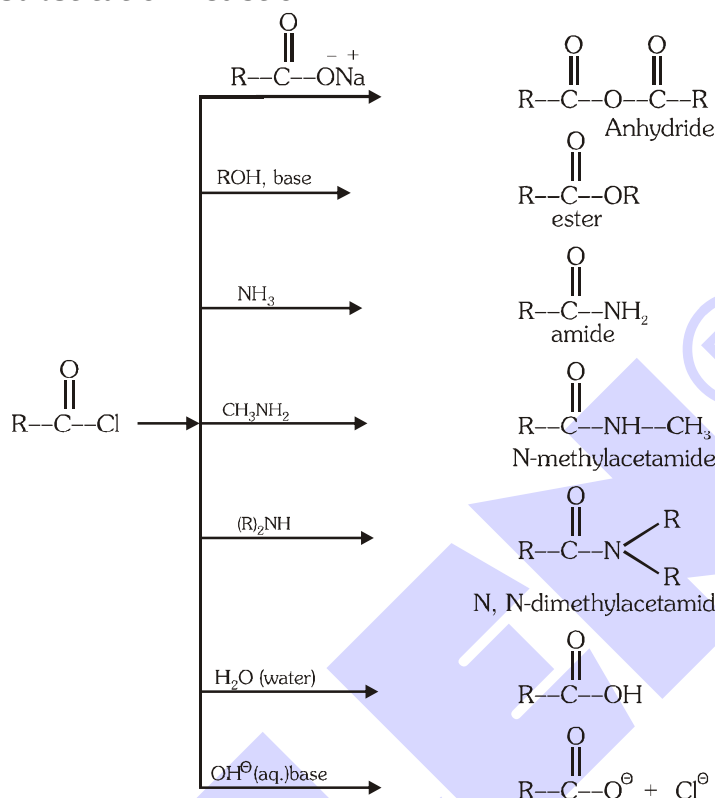
3.1 METHODS OF PREPARATION

Carboxylic acid chloride can be prepared by the reaction of carboxylic acid with PCl_5 or SOCl_2 or PCl_3 .

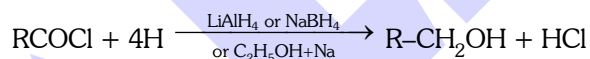


3.2 CHEMICAL PROPERTIES

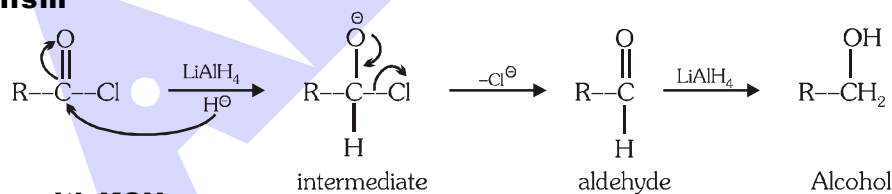
A. Nucleophilic Substitution Reaction



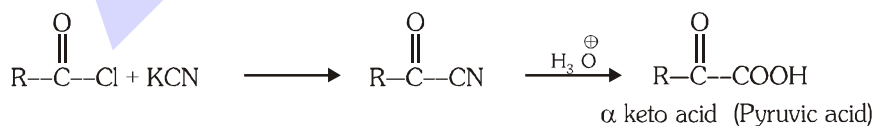
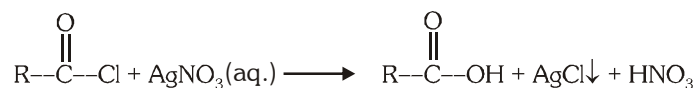
B. Catalytic hydrogenation



• Mechanism

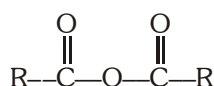


C. Reaction with KCN

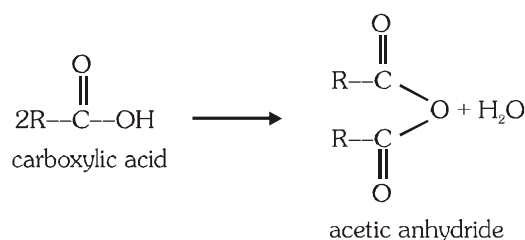
D. Reaction with AgNO₃

The aliphatic acid chlorides are readily decomposed by water therefore aqueous solution of acid chloride gives white ppt. with AgNO₃.

4.0 ACID ANHYDRIDES (RCOOCOR)



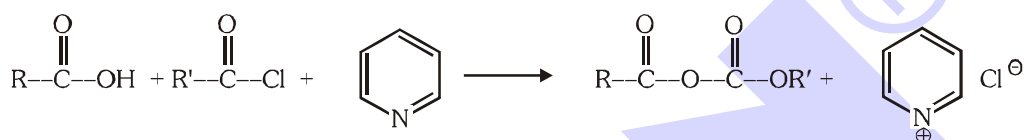
Acid anhydrides are considered to be derived from carboxylic acids by the removal of a molecule of water from two molecules of the acid.



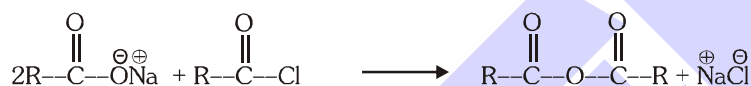
4.1 METHODS OF PREPARATION

A. Acylation

Carboxylic acid reacts with acyl chloride in the presence of pyridine to give carboxylic acid anhydride.



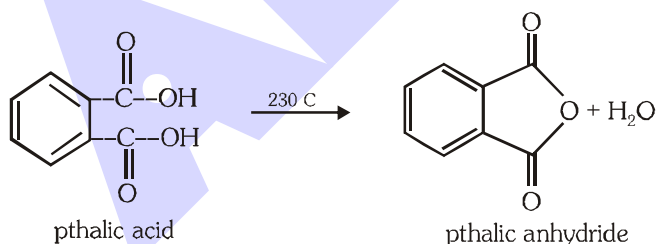
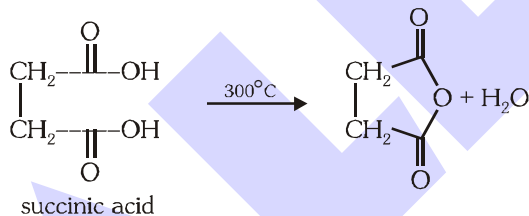
B. Reaction of acyl chloride with sodium salt of carboxylic acid



In this reaction a carboxylate ion acts as a nucleophile and brings about a nucleophilic substitution reaction at the acyl carbon of acyl chloride.

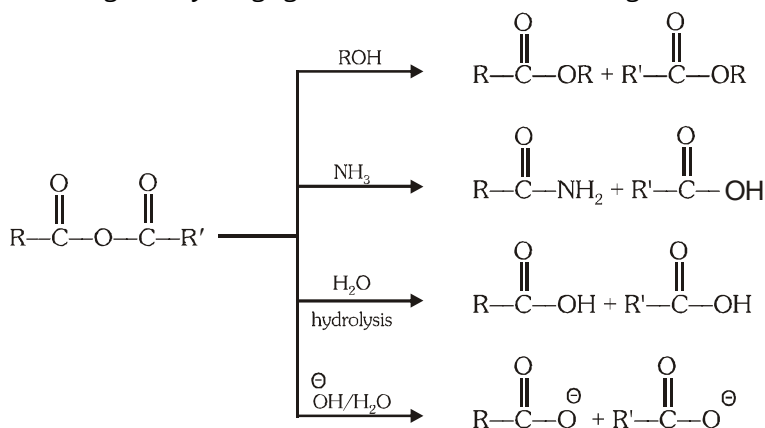
C. Cyclic anhydrides

By simple heating the appropriate dicarboxylic acid. This method leads to a five or six membered ring.



4.2 CHEMICAL PROPERTIES

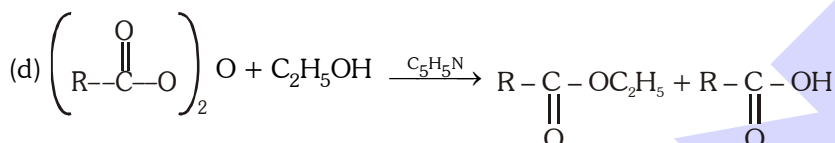
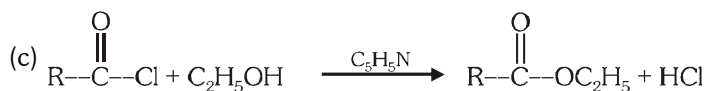
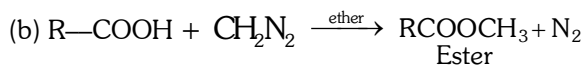
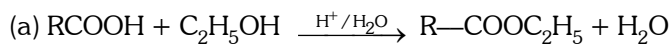
Acid anhydride are good acylating agents. Their reactions are less vigorous than the corresponding acyl halides.



5.0 ESTERS (RCOOR)

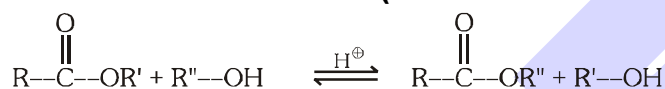
Ester are the derivative of the carboxylic acid in which the $-OH$ part of the carboxylic group has been replaced by $-OR$ group where R may be alkyl or aryl group.

5.1 METHODS OF PREPARATION

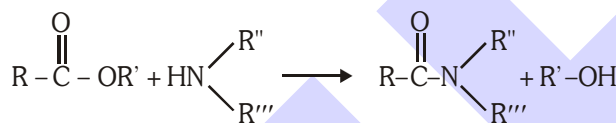


5.2 CHEMICAL PROPERTIES

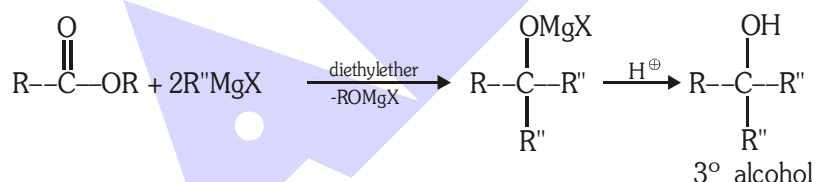
A. Conversion to other esters (Transesterifications)



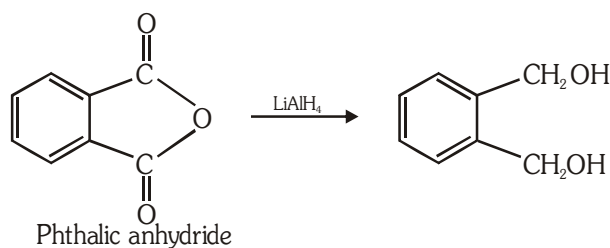
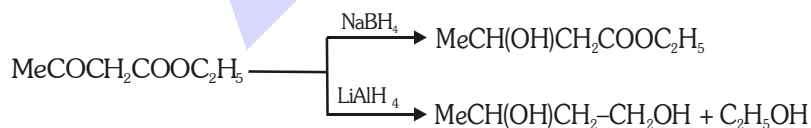
B. Conversion to amides



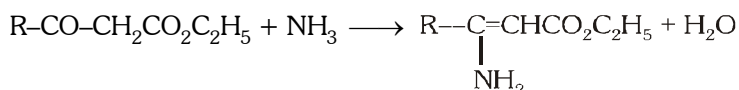
C. Reaction with Grignard Reagent



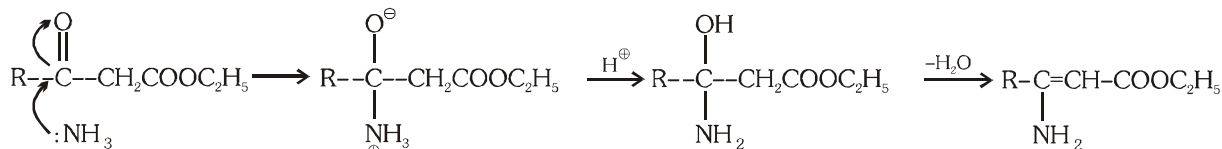
D. Reduction of ester



E. Reaction of NH₃ with keto ester

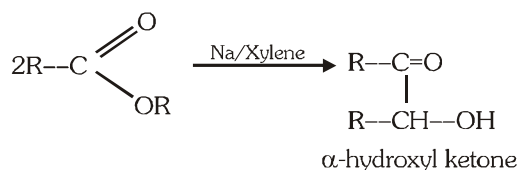


• Mechanism



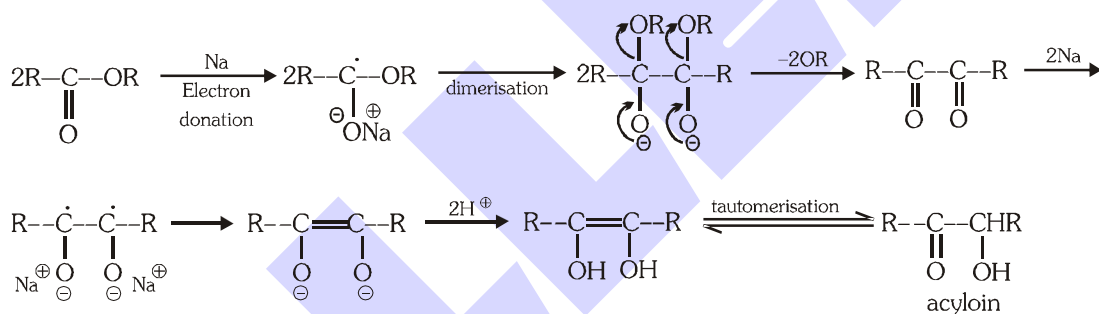
Attack will occur at carbonyl group first because of high degree of +ve charge on carbonyl carbon atom.

F. Acyloin condensation



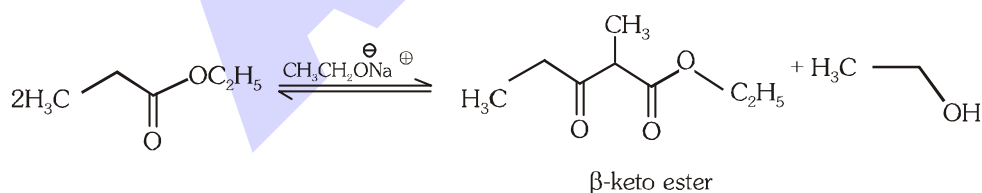
• Mechanism

It is the intermolecular, sodium promoted condensation of two moles of ester or the intra molecular condensation of a ester to α-hydroxy ketone (acyloin).



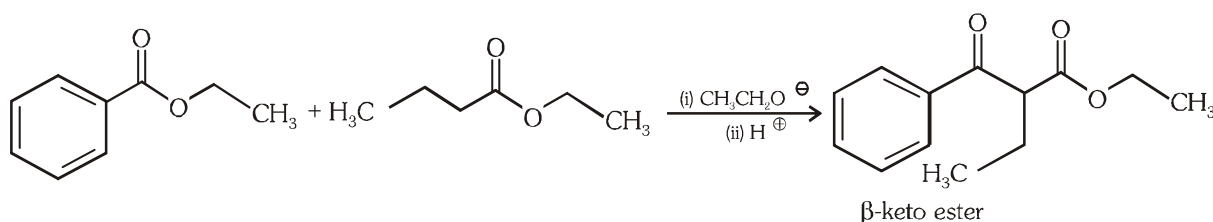
G. Claisen condensation

When two molecules of ester undergo a condensation reaction, the reaction is called Claisen condensation. The product of the claisen condensation is a β-keto ester.



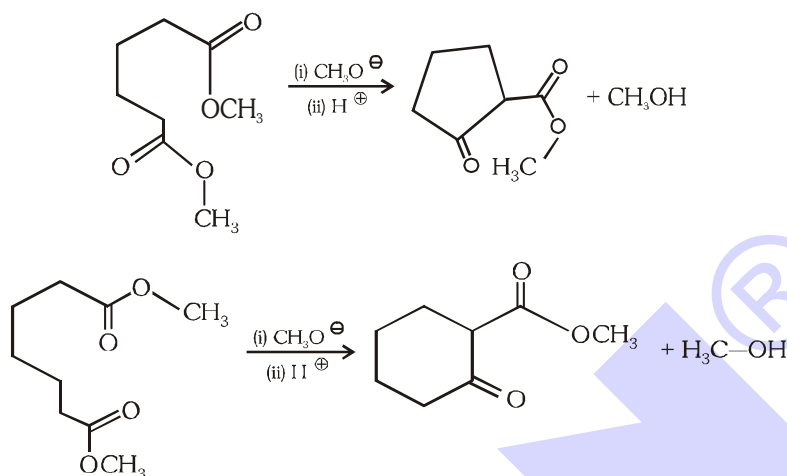
After nucleophilic attack, the aldol addition and the Claisen condensation differ. In the claisen condensation, the negatively charged oxygen reforms the carbon oxygen π-bond and eliminates the OR group.

• Mixed claisen condensation



• Intramolecular claisen condensation

Dieckmann condensation : The addition of base to a 1,6-diester causes the diester to undergo intramolecular claisen condensation, thereby forming a five membered ring β -keto ester. An intramolecular claisen condensation is called a Dieckmann condensation.

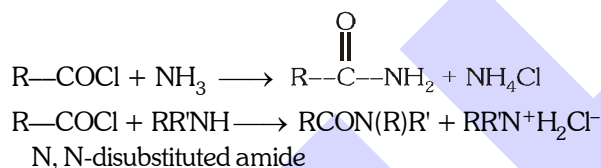


6.0 AMIDES (RCONH₂)

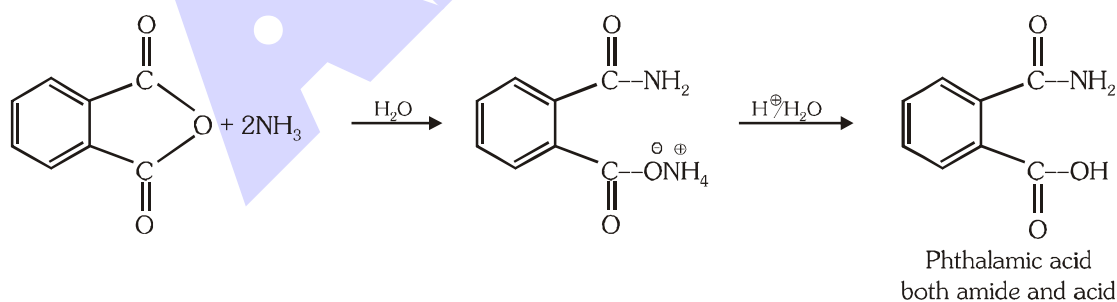
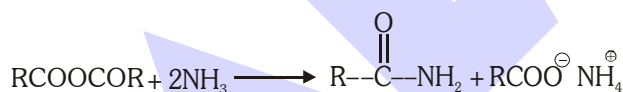
6.1 METHODS OF PREPARATION

A. Amides from Acyl chloride

Primary amines, secondary amines and ammonia all react rapidly with acid chloride to form amides. An excess of ammonia or amine is used to neutralize the HCl that would be formed otherwise.

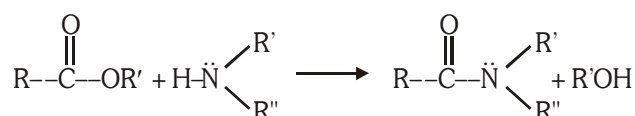


B. Amides from acid anhydride

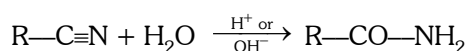


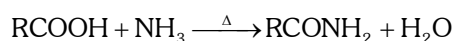
C. From Esters (Ammonolysis)

Ester undergoes nucleophilic substitution at their acyl carbon by nucleophilic ammonia or its derivative

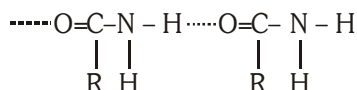


D. By partial hydrolysis of alkyl nitriles

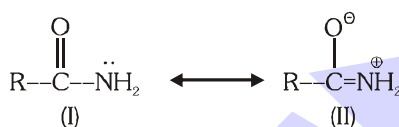


E. From Carboxylic Acid**6.2 PHYSICAL PROPERTIES**

All amides except formamide are crystalline solid at room temperature. They have relatively high melting and boiling point due to association of amide molecules by inter molecular hydrogen bonds.

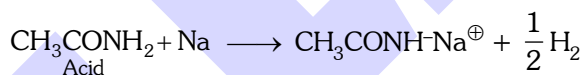
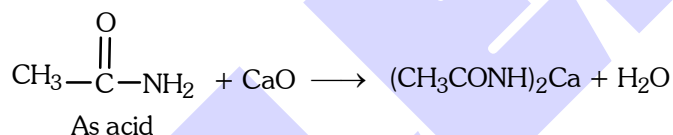
**6.3 CHEMICAL PROPERTIES****A. Amphoteric character**

Amides are very feeble bases due to involvement of pair of electron present on N atom in resonance with carboxyl groups. As a result N atom receives partial +ve charge showing feeble acidic character as well



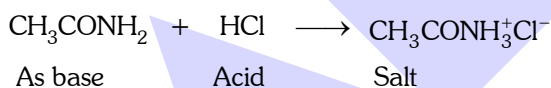
- Acidic character**

Due to structure II amide can act as acid.

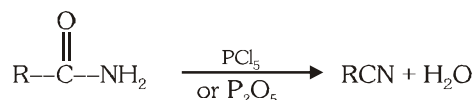
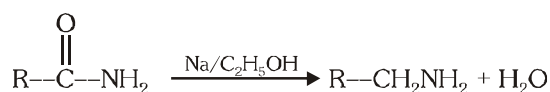


- Basic Character**

Due to structure I having lone pair of electrons on N atom it acts as base.

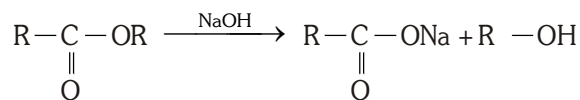
**B. Reaction with nitrous acid**

The reaction proceeds via the attack of electrophilic species NO^+ generated from HNO_2 .

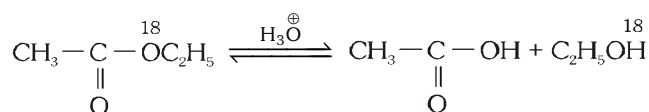
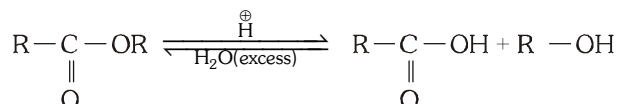
C. Reaction with PCl_5 **D. Reduction**

7.0 HYDROLYSIS OF ACYL DERIVATIVES

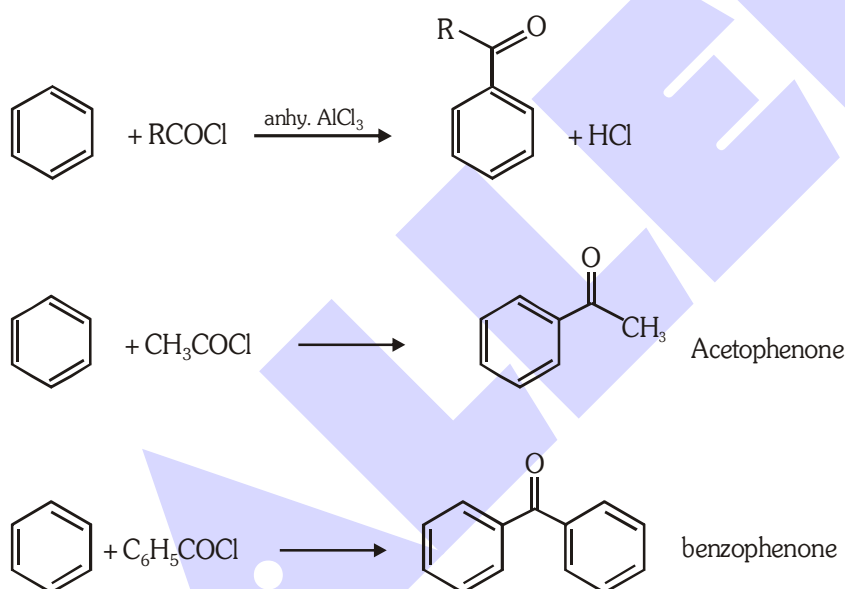
• By alkaline hydrolysis



• By acidic hydrolysis



8.0 ACYLATION OF AROMATIC COMPOUND [FRIEDEL CRAFT'S REACTION]



GOLDEN KEY POINTS

- Carboxylic acid and its derivatives (except amide) give alcohol on reduction with LiAlH₄.
- Amide gives amine on reduction with LiAlH₄.

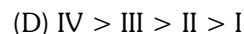
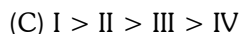
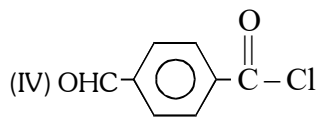
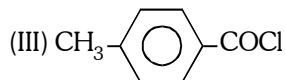
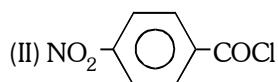
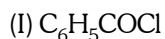
BEGINNER'S BOX-2

- The correct reactivity order with the nucleophile is :

(A) CH ₃ COCl > CH ₃ CONH ₂ > CH ₃ COOCH ₃	(B) CH ₃ COCl > CH ₃ COOCH ₃ > CH ₃ CONH ₂
(C) CH ₃ CONH ₂ > CH ₃ COOCH ₃ > CH ₃ COCl	(D) CH ₃ COOCH ₃ > CH ₃ COCl > CH ₃ CONH ₂
- Which one of the following reactions can be used for the preparation of β-hydroxy ester:

(A) Perkin reaction	(B) Reformatsky reaction
(C) Aldol condensation	(D) Claisen condensation

3. Arrange following compounds in decreasing order of reactivity for hydrolysis reaction :



4. The reaction of an amide with bromine and alkali to form a primary amine is called :-

(A) Hunsdiecker reaction

(B) Hofmann mustard oil reaction

(C) Hoffmann bromamide degradation of amides

(D) Hell-Volhard-Zelinski reaction

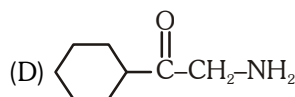
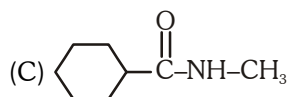
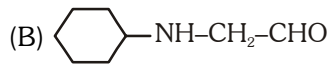
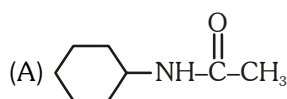
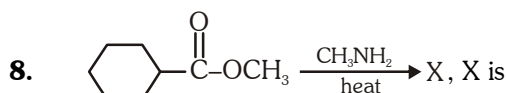
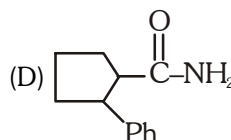
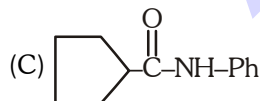
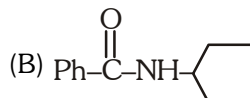
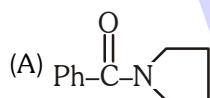
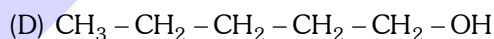
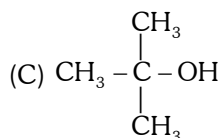
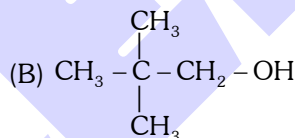
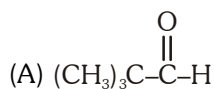
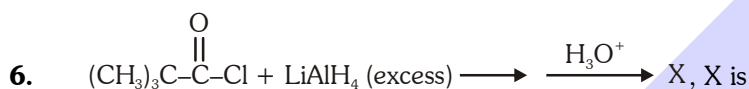
5. The treatment of an ester with LiAlH_4 followed by acid hydrolysis produces :

(A) Two aldehyde

(B) One carboxylic acid and one alcohol

(C) Two alcohols

(D) Two acids

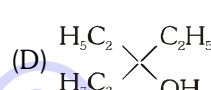
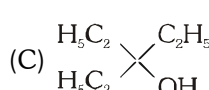
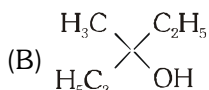
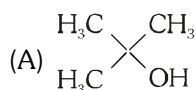


9. Which will form lactone on treatment with NaOH ?

- (A) α -Bromo acid (B) β -Bromo acid (C) β -Hydroxy acid (D) δ -Bromo acid

10. Ethyl acetate $\xrightarrow[\text{Excess}]{\text{CH}_3\text{MgBr}}$ $\xrightarrow{\text{H}_3\text{O}^+}$ P

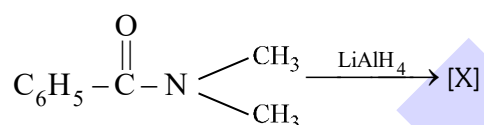
The product P will be :-



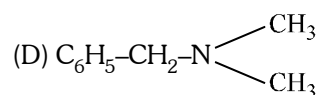
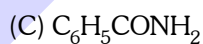
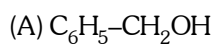
11. Acids have much higher boiling points than isomeric esters because :-

- (A) Acids form dimers by H-Bonding
(B) Acids are volatile in steam
(C) Ester are non-volatile
(D) Acids can ionise to give protons in aqueous solution

12. In the given reaction :



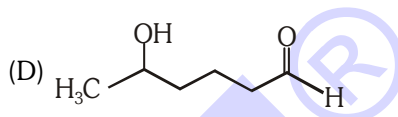
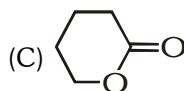
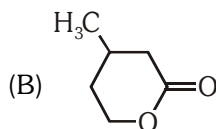
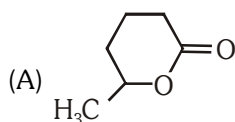
[X] will be :



SOME WORKED OUT ILLUSTRATIONS

Illustration 1.

End product of this conversion $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-(\text{CH}_2)_2-\text{CO}_2\text{H} \xrightarrow[\text{(ii) H}_2\text{O/H}^+/\Delta]{\text{(i) NaBH}_4}$ is ?



Solution (A)

NaBH_4 reduces reactant to

$\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$ which forms ester.

Illustration 2.

When acetic acid reacts with ketene, product formed is :

- (A) ethyl acetate (B) aceto-acetic ester (C) acetic anhydride (D) no reaction

Solution. (C)

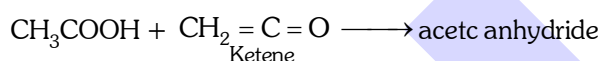


Illustration 3.

$\text{R}-\text{CH}_2-\text{CH}_2\text{OH}$ can be converted in $\text{R}-\text{CH}_2\text{CH}_2\text{COOH}$. The correct sequence of reagents is :

- (A) PBr_3 , KCN , H^+ (B) PBr_3 , KCN , H_2 (C) KCN , H^+ (D) HCN , PBr_3 , H^+

Solution (A)

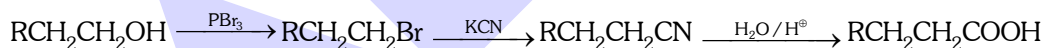
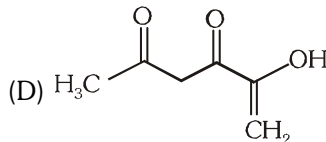
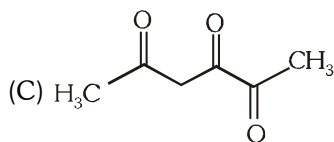
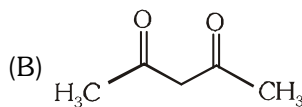
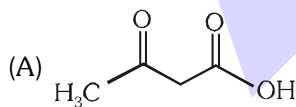
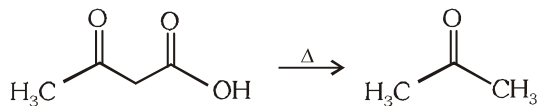


Illustration 4.

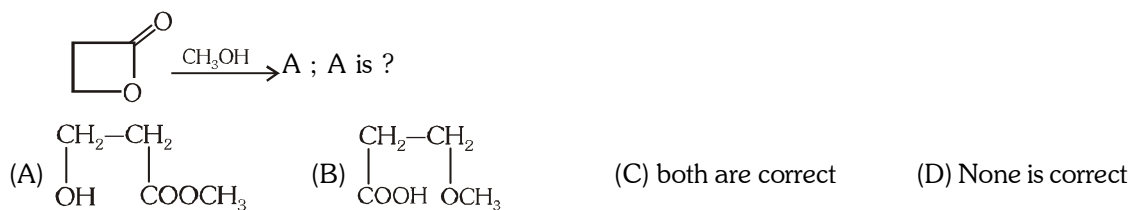
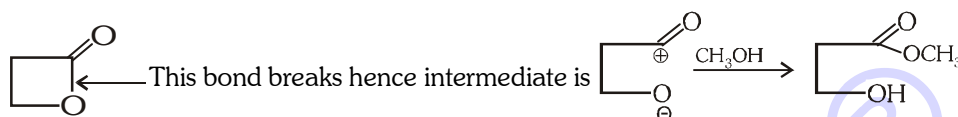
Which of the following compound would be expected to decarboxylate when heated :



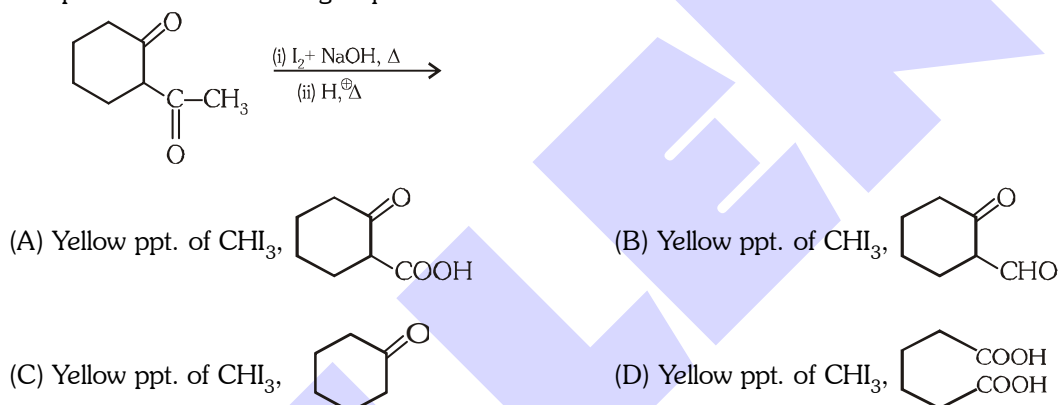
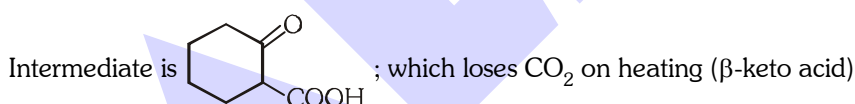
Solution. (A)



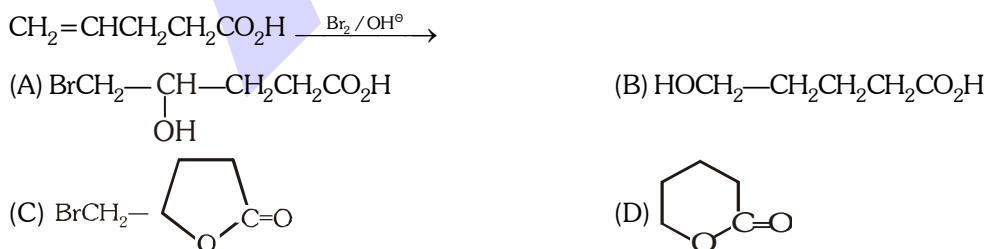
In case of β -keto acid, the decarboxylation occurs readily due to 6-membered low energy transition state formation.

Illustration 5.**Soluton (A)****Illustration 6.**

End product of the following sequence of reaction is :

**Soluton (C)****Illustration 7**

The final product obtained in the reaction :

**Solution. (C)****Illustration 8.**

On subjecting mesityl oxide to the iodoform reaction, one of the products is the sodium salt of an organic acid. Which acid is obtained ?

- (A) $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2\text{COOH}$
 (B) $(\text{CH}_3)_2\text{CH}-\text{COOH}$
- (C) $(\text{CH}_3)_2\text{C}=\text{CH}-\text{COOH}$
 (D) $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CO}-\text{COOH}$

Solution. (C)

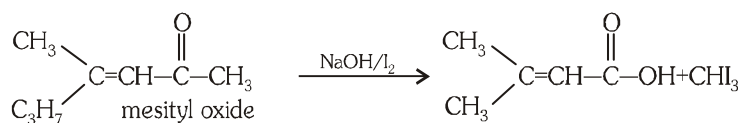
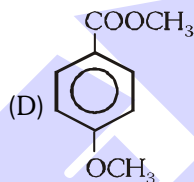
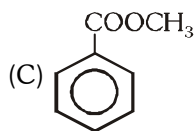
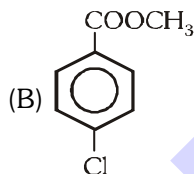
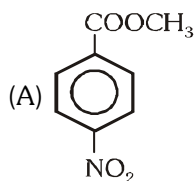
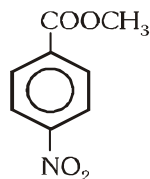


Illustration 9.

The ease of alkaline hydrolysis is more for :



Solution. (A)



There is more electron deficiency on carbonyl carbon.

Illustration 10.

Which of the following does not undergo Hell-volhard Zelinsky reaction ?

(A) HCOOH

(B) CCl₃COOH

(C) C₆H₅COOH

(D) All

Solution. (D)

None of these contain alpha H-atom.